

AI Quiz Generator

About Me

I am **Deepika Reddy**, a 3rd-year B.Tech student at Ashoka Women's Engineering College, specializing in Computer Science and Engineering with Artificial Intelligence (CSE-AI). I have a strong interest in Artificial Intelligence, Machine Learning, and Data Science, with skills in Python programming and Computer Vision.

Through this internship, I worked on an **AI Quiz Generator** project, where I gained practical experience in Python programming, file handling, answer validation, score calculation, and user interaction through terminal-based applications. This project helped me improve my problem-solving abilities and understand the practical implementation of AI-based interactive systems.

I am passionate about learning new technologies and applying AI concepts to solve real-world problems. My goal is to build a successful career in Artificial Intelligence and Data Science by continuously improving my technical and analytical skills.

Acknowledgement

I am grateful to **THINK CHAMP PV LTD** for providing me with the opportunity to complete my internship and work on the project "**AI Quiz Generator**". This internship has been a valuable learning experience that helped me gain practical exposure and improve my technical skills in Python programming and application development.

I would like to express my sincere thanks to my internship mentor for their continuous guidance, encouragement, and valuable feedback throughout the project development process.

I also extend my gratitude to the management and staff of **THINK CHAMP PV LTD** for their support, cooperation, and for providing the necessary resources to successfully complete this project.

AI Quiz Generator

Introduction

The AI Quiz Generator is a simple yet interactive Python-based application designed to conduct quizzes through the terminal or command prompt. The system acts like a basic AI assistant by communicating with users, asking questions, accepting answers, validating responses, calculating scores, and displaying the final results automatically.

The main purpose of this project is to create an easy-to-use quiz platform that demonstrates the practical implementation of Python programming concepts such as loops, conditions, file handling, lists, and user interaction. The application reads quiz questions from a text file, presents them to the user one by one, checks whether the entered answers are correct or not, and updates the score accordingly.

The project provides a user-friendly and interactive environment where users can test their knowledge in different subjects. At the end of the quiz, the system displays the final score and stores the result in a text file for future reference. The project can also be enhanced with advanced features such as timers, multiple-choice questions, difficulty levels, random question generation, and colored outputs.

Project Objective

The primary objective of the AI Quiz Generator project is to develop a Python application that can automatically conduct quizzes and evaluate user performance. The project focuses on creating an interactive system that communicates with users and provides instant feedback based on their answers.

The objectives of the project are:

- To design a quiz application using Python programming language.
- To create an interactive system that asks quiz questions to users.
- To accept answers from users through the terminal.
- To validate user answers by comparing them with correct answers.
- To calculate the total score obtained by the user.
- To display the final quiz result after completion.
- To store user scores permanently in a text file.
- To improve programming skills using real-time project development.

The project also aims to provide practical exposure to software development and enhance logical thinking and problem-solving abilities.

Project Description

The AI Quiz Generator is a terminal-based application developed using Python. The application works by reading quiz questions and answers from a text file. Each question is displayed to the user one at a time, and the user enters the answer using the keyboard.

The system then compares the entered answer with the correct answer stored in the file. If the answer is correct, the score increases. Otherwise, the system displays the correct answer to the user. This process continues until all questions are completed.

At the end of the quiz, the application displays the final score along with the total number of questions attempted. The score is also saved automatically in a text file called “scores.txt” so that records can be maintained for future reference.

The application uses Python modules such as:

- random → to shuffle questions randomly
- file handling → to read and write files
- loops and conditions → to manage quiz flow and answer validation

The project is simple, efficient, and beginner-friendly. It demonstrates how Python can be used to create interactive command-line applications.

Mandatory Features

1. Welcome Message

When the program starts, the application displays a welcome message such as “Welcome to AI Quiz Generator” to greet the user and provide a better user experience.

2. Minimum 5 Quiz Questions

The application contains at least five quiz questions stored in a text file. These questions are asked one by one during the quiz session.

3. User Answer Input

The system allows users to type their answers through the keyboard using the input() function in Python.

4. Answer Validation

After receiving the answer from the user, the application checks whether the answer is correct or incorrect by comparing it with the stored answer.

5. Score Calculation

The system automatically calculates the score. For every correct answer, the score is increased by one.

6. Loop for Multiple Questions

The application uses loops to display multiple quiz questions continuously until all questions are completed.

7. Final Result Display

After completing the quiz, the system displays the final score in the format:
Final Score: 4/5

8. Save Score in Text File

The final score is stored in a text file named “scores.txt” using Python file handling techniques. This helps in maintaining quiz records permanently.

Project Structure

```
AI_Quiz_Generator/  
├── quiz.py  
├── questions.txt  
├── scores.txt  
└── README.md
```

Python Source Code

```
import random  
  
# Welcome Message  
  
print("=====")  
  
print(" Welcome to AI Quiz Generator")  
  
print("=====\n")  
  
score = 0  
  
# Read questions from file  
  
with open("questions.txt", "r") as file:  
    lines = file.readlines()
```

```
questions = []

# Convert data into question-answer pairs
for line in lines:

    question, answer = line.strip().split("|")

    questions.append((question, answer))

# Shuffle questions randomly
random.shuffle(questions)

# Ask questions one by one
for i, (question, answer) in enumerate(questions, start=1):

    print(f"Question {i}:")

    print(question)

    user_answer = input("Your Answer: ").strip().lower()

    # Validate answer
    if user_answer == answer.lower():

        print("Correct Answer!\n")

        score += 1

    else:

        print("Wrong Answer!")

        print("Correct Answer:", answer, "\n")

# Display final result
print("=====")

print(f"Final Score: {score}/{len(questions)}")

print("=====")

# Save score in text file
with open("scores.txt", "a") as file:
```

```
file.write(f"Score: {score}/{len(questions)}\n")  
  
print("Score saved successfully!")
```

Questions File

What is Python?|programming language
What is AI?|artificial intelligence
Who developed Python?|guido van rossum
What does CPU stand for?|central processing unit
What is the full form of RAM?|random access memory

Sample Output

Welcome to AI Quiz Generator

Question 1:
What is Python?
Your Answer: programming language
Correct Answer!

Question 2:
What is AI?
Your Answer: artificial intelligence
Correct Answer!

Final Score: 2/2

Tools Used

- Python
- VS Code / PyCharm

Concepts Used

- Loops
- Conditions
- File Handling
- Lists
- User Input

Conclusion

The project successfully demonstrates the implementation of a Python-based quiz application with answer validation, score calculation, and file handling.

